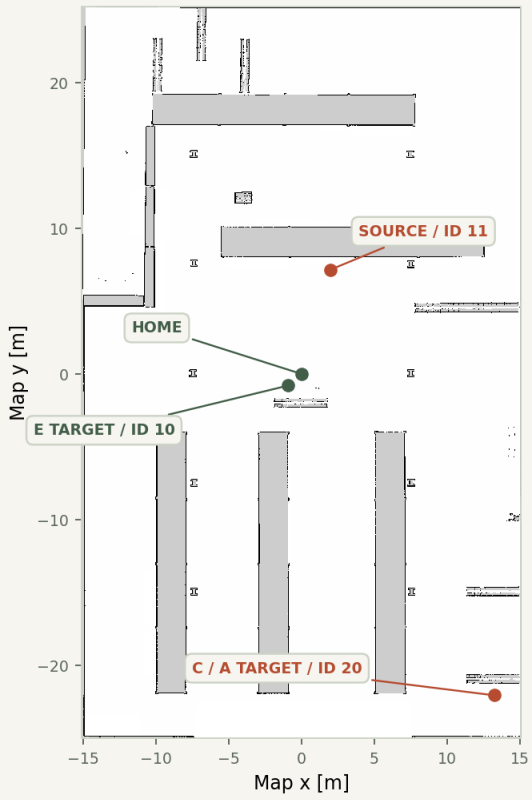
Warehouse mobile manipulation.

Nithyasri Rajulu | Robotics course project

A ROS 2 project connecting a mobile robot, an articulated arm and a vacuum gripper. The task centres on collecting an object from a known warehouse location and placing it at a selected destination.



MAP → TASK

Known standing poses

Home
x 0.00 / y 0.00 m

Source · ID 11
x 1.98 / y 7.15 m

E target · ID 10
x −0.90 / y −0.80 m

C / A target · ID 20
x 13.24 / y −22.07 m

Configured positions
No recorded trajectory

SUPPLIED OCCUPANCY MAP / CONFIGURED TASK POSITIONS

Engineering focus

The project brings together task coordination, mobile navigation, coordinate transforms, arm kinematics and robot-control interfaces within a course-provided simulation framework.

Tools: ROS 2 Jazzy, Nav2, Gazebo, py_trees and Python.

Connecting the system.

Task decisions connect to navigation and manipulation interfaces.

Mission coordination

A Python mission node selects the task mode and coordinates navigation, arm motion and vacuum commands. The task logic includes sequential and behaviour-tree approaches.

Mobile navigation

Nav2 actions provide goal-based motion between configured standing poses. The navigation configuration combines a global planner, a path-following controller and obstacle costmaps.

Arm and gripper

A URDF-derived model describes the arm kinematics. Joint-trajectory actions request arm motion, while attach/detach commands operate the vacuum-gripper interface.

Localization and feedback

Robot pose, odometry, TF transforms and joint states connect the components. The advanced course mode introduces AMCL localization and a spin-to-localize behaviour.

From a task to tool motion.

Project approach and software structure

Mission sequence

- Undock and navigate to the source standing pose.
- Approach the object, request vacuum attachment and lift the arm.
- Navigate to the destination and request placement.

Kinematic approach

Forward kinematics composes the joint transforms from the arm base to the tool. Target positions are converted from the map frame into the arm frame using TF. Numerical inverse kinematics searches for a joint configuration corresponding to the requested tool pose.

Behaviour-tree organisation

Sequences organise task stages, while selectors choose between an existing condition and an associated action. Navigation, localization and manipulation are expressed through reusable task behaviours.

Project scope
This overview describes the supplied course implementation and its engineering approach. The task uses configured warehouse positions. It is presented as a simulation-development project within the supplied framework.